

Mehedi Hasan

✉ mmehedihasan29088@gmail.com 🌐 mehedihasan88.github.io/portfolio/ 🔄 mehedihasan88 📄 mehedi-hasan88

I am interested in developing trustworthy, interpretable, and human-centered intelligent systems. My research focuses on artificial intelligence for software engineering, multimodal learning, and human-computer interaction, with an emphasis on reliability, transparency, and real-world applicability.

EDUCATION

Jahangirnagar University

BSc. in Computer Science & Engineering

Feb 2018 – July 2023

GPA: 3.47/4.0

- Thesis: A Machine Learning-based Image Forgery Detection.
- Coursework: ML, NLP, Image processing, OS, DSA, Graph Theory, ToC, SE, OOP, SAD, Compiler Design.

EXPERIENCE

Samsung R&D Institute Bangladesh

Software Engineer

Dhaka, Bangladesh

Jan 2024 – Present

Innovation & Research | Key Field: ML, NLP, Agentic AI, XAI, Wearable devices

- **Developed an interactive prior-art search agent that refines user queries and detects multi-patent obviousness relationships with explanations**; selected for AI-PoC exhibition and currently being integrated into Samsung Research Bangladesh's in-house patent search workflow to streamline office-wide innovation processes.
- Proposed initial idea and contributed to the development of a **novel machine learning based methodology for liquid consumption estimation using wearable devices**.
- Actively contribute to team innovation meetings by evaluating peer-submitted ideas, providing constructive technical feedback on novelty, obviousness, and feasibility.

Development | Tech Stack: JAVA, Kotlin, Jetpack Compose, Android, HILT, WearOS

- Worked on Samsung Galaxy Wearable projects, developing core modules (notifications, nowbar, watchface overlay), optimizing performance, and prototyping new features for Galaxy Watch 7 and 8.
- Designed and built the complete backend module for the ongoing activity manager (Nowbar) using concurrent data structures
- Developed nowbar overlay module integrating Google's AOD interface and Samsung's new Low Power AOD chip.
- Migrated MVP-based settings module from XML to Jetpack Compose-based MVVM architecture reducing load time (30ms to 5ms).
- Developed a POC for link preview system for the watch notification panel, identifying security vulnerabilities and solutions.
- Built a backup and restore (BnR) module to preserve user settings while changing/connecting new devices.
- Delivered high-quality fixes for 175+ QA/market-reported issues with team lowest rejection rate.

PUBLICATIONS

- **Mehedi Hasan, Mahbub Islam Mahim, Jugal Krishna Das (2026). Segmentation Strategies for Long-Code Authorship Detection: Balancing Context Coverage and Computational Efficiency.** | *Under Review*
 - Compared truncation, multi-view cropping, and sliding-window segmentation strategies for long-code authorship detection under transformer context limits. Sliding-window achieved the best performance, while multi-view reached near-parity with far fewer computations, highlighting context coverage as the key driver of accuracy.
- **Mehedi Hasan, Mahbub Islam Mahim (2025). Divide and Conquer: A Two-Stage Label-Wise Cascaded Framework for Bangla Hate Speech Detection.** | [Codes](#) [Paper](#)
 - Developed a cascaded pipeline combining binary and multiclass models with k-fold cross-validation and intra-label ensembling; achieved a 0.7392 micro-F1 score — the top performance in the BLP Workshop 2025 Shared Task, awarded **Best Paper**, and presented at ACL-IJCNLP 2025.

PROJECTS

- **Human-Machine-Hybrid-Adversarial-Code-Classification** [🔗](#) – *PyTorch, scikit-learn, NumPy, Pandas*
 - Implemented multiple strategies, including AST-based feature engineering, class-mapping (4-to-3), overlapping window analysis for long code, multi-view learning, and specialized hard-case training loops.
 - Achieved macro F1 of 82% on seen languages and 62% on mixed (seen and unseen languages) setups.
- **Image Forgery Detection** [🔗](#) – *OpenCV, NumPy, SciPy, scikit-learn, Matplotlib, Seaborn*
 - Copy-move and Splicing image manipulation identification using LBP, DCT and SVM.
 - Best setup achieved up to 97% accuracy on CASIA datasets and 89% on COLUMBIA for forgery detection.

COMPETITIONS & ACHIEVEMENTS

Competitive programming & Problems solving

- Solved **1,900+ algorithmic problems** across multiple online judges, participated at **160+ contests**.
- Achieved highest rating of **1611 (Expert)** on Codeforces (@MhD88) [🔗](#) and **1730 (3-star)** on CodeChef (@mhd88) [🔗](#).
- Top Contest Performances: 40th among 477 teams (@bit_matrix) [🔗](#), Top 2% among 12,275 participants (@MhD88) [🔗](#), Top 2% among 16,010 participants (@MhD88) [🔗](#)

Hackathons

- Became **Finalist among 400+** teams (Team: JU_NONAME) at Code Samurai Hackathon 2022, hosted by University of Dhaka. Solved algorithmic and SQL problems in preliminaries; developed a web-based system in the final. [🔗](#)

ML Contests & Shared Tasks

- Ranked **1st among 90 teams** (Team: team_hate_mate) at the BLP Workshop 2025 Shared Task [🔗](#)

SCHOLARSHIPS & AWARDS

- Received **Best Shared Task Paper Award** at Second Workshop on Bangla Language Processing (Dec 2025) [🔗](#)
- Received **Excellence Award** for development contributions at Samsung during Jan–Mar 2024. [🔗](#)
- Received an award for achieving the professional-level Software Coding (SWC) certification test on November 08, 2024, in recognition of coding excellence. [🔗](#)
- Received **Dean's Merit scholarships** in all four years of B.Sc. for academic excellence (2018–2023) [🔗](#)
- Merit Scholarship in Junior School Certification Examination, DSHE, Ministry of Education (2013)
- **Talent Pool Scholarship** in Primary School Certification Exam, Ministry of Primary and Mass Education. (2011)

TEACHING & MENTORING EXPERIENCE

Competitive Programming Mentor | *Department of CSE, Jahangirnagar University* 2022 - 2023

- Provided training on C/C++, Data Structures, and Algorithms, arranged practice contests.

Academic Tutoring | *Private Tutor* 2020 - 2024

- Provided mentorship to **80+ students** on physics, mathematics and various programming disciplines.

SKILLS

Programming Languages	: C, C++, Java, Kotlin, Python, JavaScript
ML & AI	: scikit-learn, NumPy, OpenCV, PyTorch, Hugging Face Transformers, LLMs
Frameworks & Technologies	: Android, Node.js, Jetpack Compose, HILT, Bootstrap
Databases	: MySQL, MongoDB
Tools	: Git, Perforce, VS Code, Android Studio